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The impact of digital financial inclusion on equity of opportunity in education

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ABSTRACT

Under the overarching goal of building a comprehensive education powerhouse, the technological empowerment and inclusive nature of digital finance have injected new momentum into enhancing educational quality and efficiency. This study employs provincial panel data from China (2013–2023) and constructs an educational Gini coefficient to empirically examine the impact of digital inclusive finance on educational equity through a two-way fixed effects model. The findings reveal that the development of digital inclusive finance significantly promotes equitable access to education in China, with its effects moderated by regional economic development levels and demographic factors. Furthermore, digital inclusive finance indirectly enhances educational equity through improved internet penetration rates.

1. Introduction

Empowered by digital technology, digital inclusive finance is reshaping the pattern of financial services and becoming a new engine for high-quality economic development. 2023 The State Council's Implementing Opinions on Promoting the High-Quality Development of Inclusive Finance in October pointed out that the development of digital inclusive finance should be pushed forward in an orderly manner, with a focus on strengthening the empowerment of fintech to achieve a balance between innovation and risk prevention and control. This new financial form not only breaks through the time and space limitations of traditional services but also enhances the breadth and depth of service coverage of inclusive finance, bringing revolutionary changes to the accessibility of financial services. This innovative financial model overcomes traditional limitations through three key pathways: Mobile payments and online credit eliminate reliance on physical branches, enabling financial service access in remote areas (Gong et al., 2023); Big-data-driven automation replaces manual reviews, reducing operational costs and driving service fees down by over 50 % (Gong et al., 2025); Vertical innovations like education installment plans and student loans directly facilitate educational resource acquisition (Xiang et al., 2025). As a strategic pillar of national development, the allocation of resources and the quality of education development are also related to social equity and future competitiveness (Beck et al., 2018). Studies have shown that from the financial sector to the education sector, the technological advantages and conceptual value of digital inclusive finance can provide new ideas for solving the problems of imbalance of educational resources and inefficient allocation of funds, The inequality in education has shifted from a lack of enrollment opportunities to differences in access to high-quality resources in China, and promote educational equity and quality improvement (Menberu, W. A., 2024). It can be seen that exploring effective ways of digital financial empowerment for high-quality education development has become an important issue in the current reform and development of the education sector.

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Digital inclusive finance is a new type of financial model that realizes the inclusion, fairness, and progress of financial services, which has been rapidly developed and deeply integrated into the lives of residents, significantly enhancing their consumption (Borsi et al., 2022). However, in education, there is a certain degree of relativity in the equitable development of education due to factors such as differences in resource endowment and imbalanced regional economic development, and the imbalance of educational development inevitably brings about educational inequality (Carneiro and Heckman, 2002). More precisely, inequality of access to education in the era of scarcity of educational opportunities is gradually being replaced by inequality of access to quality educational resources (Ferreira, 2001). To realize the equity of educational opportunities to a greater extent, scholars at home and abroad often use the education Gini coefficient as an indicator for research, which usually includes the number of allocation of resources, the number of educated population, the number of years of education, and other indicators (Thomas, V., 2000), which has a unique advantage in reflecting the balance of the distribution of educational resources, and has gradually become the mainstream metric in the current study of educational equity.

In summary, current studies have mainly explored the impact of finance on education development from a traditional perspective, but few scholars have established a link between digital inclusive finance and education Gini coefficient. Therefore, we further analyze the impact of digital inclusive finance on educational opportunity equity based on the prequel study, and use the digital inclusive finance index to reveal the mechanism of action, summarize the heterogeneity law of digital inclusive finance on educational opportunity equity, and enrich the concluding research on the factors affecting high opportunity equity in education.

2. Theoretical analysis and hypothesis

2.1. Analysis of digital financial inclusion and equity of educational opportunities

Digital inclusive finance plays an important role in promoting economic development and social equity for disadvantaged groups. From the perspective of economic development, digital finance can promote economic growth in general (Zhang et al., 2019), and in particular, it can help low-income groups such as rural laborers, self-employed people and small and micro-enterprises (Gopal and Schnabl, 2022) to achieve income growth and alleviate the gap between the rich and the poor. When the economic gap is gradually narrowed with the help of digital inclusive finance, the economic situation of disadvantaged groups is improved, which creates more favorable conditions for the realization of educational equity. Equity in educational opportunities is an important foundation of educational equity, and the economic status of families largely affects access to educational opportunities.

As digital inclusive finance expands the service scope and contact population of traditional inclusive finance (Heijdraet et al., 2017), it effectively makes up for the coverage gap of traditional finance (Jung and Cha, 2020), prompting more funds to flow to areas in need of education funding, effectively improving the economic situation of families, so that the children of disadvantaged groups can have a fairer opportunity to obtain a fairer opportunity at the start of their education, which is important for alleviating the This is of positive significance in alleviating the uneven distribution of educational resources and promoting social class mobility (Yue et al., 2022).

Based on the above analysis, this paper proposes Hypothesis H1: Digital financial inclusion can significantly promote equity in educational opportunities.

2.2. Digital financial inclusion, internet penetration, and equity in educational opportunities

Digital financial inclusion is a product of the deep integration of traditional finance and digital technology, which relies on digital technology to break the time and space limitations of traditional financial services, and the Internet technology plays a crucial role in it. In the past decade, China's Internet penetration rate has rapidly risen from 42.1 % in 2012 to 75.6 % in 2022, and the replacement of traditional fixed networking devices by mobile Internet has gradually broadened the Internet use channels of family members (Samsudeen and Selvaratnam, 2019). With the rapid popularization of the Internet, financial technologies such as e-payment, online banking, and online lending have gradually penetrated into households, making it easier for low-income people to obtain loans and expanding the scope of financial services (Roemer and Trannoy, 2016), and at the same time, relying on the ever-increasing penetration of the Internet, financial technology has been able to more widely cover different income groups, allowing households to access digital inclusive finance to support education. In addition, the integration of information technology, such as the Internet, has changed people's access to knowledge and information, bringing new opportunities to promote educational equity (Paunov and Rollo, 2016). Specifically, that is, digital technology has broken the dilemma of traditional education constrained by factors such as geography, resources and time, and promoted the distribution of educational resources in the direction of equalization, which is of great practical significance for promoting the sharing of educational resources as well as guaranteeing fairness in educational opportunities.

Based on the above analysis, this paper proposes hypothesis H2: Digital financial inclusion can promote equity in educational opportunities by increasing Internet penetration as a whole.

3. Research design

3.1. Model setting and interpretation of variables

Summarizing the above analysis, to study the impact of digital financial inclusion on equity of educational opportunities, this paper constructs a model(1) as follows:

$$Jini_{i,t} = \beta_0 + \beta_1 \cdot Findex_{i,t} + \mu_{i,t} + \alpha_i + \alpha_t + \varepsilon_{i,t} \quad (1)$$

3.2. Sample selection and data sources

In this paper, the panel data at the provincial level from 2013 to 2023 are selected as the research object, and the raw data are cleaned and processed, and analyzed by Stata. The data of explanatory variables in this paper come from the Peking University Digital Inclusive Finance Index, while the explanatory variables, mediating variables, and control variables come from the China Statistical Yearbook, China Education Statistical Yearbook, and China Tertiary Industry Statistical Yearbook.

3.3. Descriptive statistics

Specific variable definitions are provided in Appendix A. Some of the variables in this paper are standardized and logarithmic to ensure comparability and robustness of the findings while also controlling for time and individual fixed effects.

The results of descriptive statistics are shown in Table 1. As can be seen from the table, the mean value of the education Gini coefficient is 0.226, the minimum value is 0.160, the maximum value is 0.530, and the standard deviation is 0.052, indicating that there is a certain imbalance in education opportunities in various regions. The mean value of the digital financial inclusion index is 5.615, the standard deviation is 0.319, and the median is higher than the mean, indicating that the level of digital financial inclusion development in most regions is higher than the national average. The development differences between areas are more significant.

4. Empirical tests

4.1. Benchmark regression analysis

Column (1) of Table 2 reports the results of the baseline regression on equity in educational opportunities in the digital inclusion regions. The regression coefficient of the core explanatory variable Findex is −0.062 and significantly negative at the 1 % level after adding all control variables and fixing for region and time effects. It indicates that developing digital financial inclusion can significantly reduce the education Gini coefficient and promote equity in educational opportunities. The internal logic may lie in the fact that the higher the level of development of digital inclusive finance, the easier it is for low-income regions to access financial services, increase household disposable income, reduce the regional distribution differences in educational resources, and alleviate educational inequality. Indeed, as China continues to develop its digital inclusive finance system, the corresponding monetary policy has fully addressed educational equity concerns. It maximizes support for basic education development, thereby easing public concerns about educational fairness. Hypothesis 1 of this paper is verified.

4.2. Robustness tests

4.2.1. Endogeneity test

The level of regional digital inclusive finance development will significantly reduce the inequality of educational opportunities, and in turn, the increase in equity of educational opportunities may further promote the level of digital inclusive finance development, and there may be a bidirectional causal relationship between the two. To effectively address the endogeneity issue, this study selects the number of regional Internet access ports as an instrumental variable to be tested. Internet infrastructure is a prerequisite for the development of digital finance, and equity in educational opportunities is mainly affected by institutional factors such as household registration policy and financial education expenditure, which are not directly related to Internet infrastructure and satisfy the exclusivity constraint. Columns (4)(5) of Table 2 show the regression results after using the number of Internet access ports as an instrumental variable, the coefficient of Findex is −0.235, which is negatively significant at the 1 % level, and the F-statistic of the weak instrumental variable test is 18.93, which is greater than 10, rejecting the original hypothesis that the instrumental variable is a weak instrumental variable. The coefficient of Findex remains significant after using instrumental variables, supporting the robustness

Table 1
Descriptive statistics.

VarName	Obs	Mean	SD	Min	Median	Max
Jini	341	0.226	0.052	0.160	0.216	0.530
Findex	341	5.615	0.319	4.746	5.683	6.118
Gdp	341	5.319	0.978	2.099	5.456	7.213
Reg	341	60.944	12.277	23.930	60.080	89.600
Sav	341	0.292	0.055	0.106	0.293	0.449
Popgrw	341	3.444	3.851	−6.920	3.730	11.470
Inrat	341	2.500	0.366	1.794	2.435	3.556
Fisc	341	0.435	0.183	0.056	0.401	0.894
Cover	341	5.539	0.387	4.305	5.620	6.145
Port	341	7.501	0.960	3.786	7.642	9.249
Ipr	310	4.035	0.230	3.484	4.069	4.521

Table 2
Digital financial inclusion and equity in educational opportunities.

VARIABLES	(1) Jini	(2) Jini	(3) Jini	(4) IV-First	(5) IV-Second
Findex	−0.062***				−0.235**
	(0.020)				(0.097)
L.Findex		−0.087***			
		(0.019)			
Cover			−0.073***		
			(0.014)		
Port				0.061***	
				(0.014)	
Gdp	−0.026***	−0.024***	−0.028***	−0.003	−0.025***
	(0.005)	(0.005)	(0.005)	(0.013)	(0.005)
Reg	0.002***	0.002***	0.002***	0.012***	0.004***
	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)
Sav	−0.018	−0.014	−0.026	−0.048	−0.028
	(0.020)	(0.021)	(0.019)	(0.074)	(0.023)
Popgrw	−0.001***	−0.001***	−0.001***	−0.000	−0.001**
	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)
Inrat	0.004	0.007	−0.002	0.037	0.009
	(0.009)	(0.010)	(0.009)	(0.026)	(0.009)
Fisc	0.024	0.022	0.019	0.009	0.030*
	(0.016)	(0.016)	(0.016)	(0.042)	(0.017)
Constant	0.569***	0.673***	0.582***	3.840***	1.187***
	(0.094)	(0.089)	(0.067)	(0.184)	(0.423)
Observations	341	310	341	341	341
R-squared	0.370	0.410	0.404	0.996	0.973
Year FE	YES	YES	YES	YES	YES
ID FE	YES	YES	YES	YES	YES

Note: Standard errors in parentheses.
*, ** and *** indicate significance at the 10 %, 5 % and 1 % levels, respectively.

of the results.

4.2.2. Other robustness tests

(1) One-period lagged regression

Considering that there is a certain lag in the impact of the development of digital financial inclusion on the promotion of equity in educational opportunities, the independent variable digital financial inclusion index is lagged by one period, and the regression results are shown in Column (2) of Table 2, where the lagged one-period digital financial inclusion index is significantly negative at the 1 %

Table 3
Mechanism test analysis.

VARIABLES	(1) Jini	(2) Ipr
Findex	−0.062***	0.534***
	(0.020)	(0.174)
Gdp	−0.026***	−0.088**
	(0.005)	(0.043)
Reg	0.002***	0.024***
	(0.000)	(0.004)
Sav	−0.018	−1.072***
	(0.020)	(0.181)
Popgrw	−0.001***	0.003
	(0.000)	(0.004)
Inrat	0.004	−0.083
	(0.009)	(0.089)
Fisc	0.024	−0.045
	(0.016)	(0.149)
Constant	0.569***	0.785
	(0.094)	(0.825)
Observations	341	310
R-squared	0.370	0.879
Year FE	YES	YES
ID FE	YES	YES

Note: Standard errors in parentheses.
*, ** and *** indicate significance at the 10 %, 5 % and 1 % levels, respectively.

level, which is consistent with the benchmark regression results.

(2) Replacement of explanatory variables

The core explanatory variable is further replaced with the breadth of digital financial coverage, an index that quantifies the spatial reach of digital financial services through indicators such as account coverage (Guo et al., 2024). The regression results, as shown in column (3) of Table 2, show that the breadth of digital financial coverage is still significantly negative at the 1 % level, which further validates the robustness of the results.

4.3. Mechanism analysis

In order to explore the transmission path of the impact of digital financial inclusion on the equity of educational opportunities, with reference to the inference idea ofZhang et al. (2021), this paper selects the Internet penetration rate as the mediating variable to be tested and constructs the following model(2).

$$IPR_{i,t} = \beta_0 + \beta_1 \cdot Findex_{i,t} + \mu_{i,t} + \alpha_i + \alpha_t + \varepsilon_{i,t} \tag{2}$$

The regression results are shown in Table 3. Column (1) reports the results of the baseline regression on the impact of digital financial inclusion on equity in educational opportunities. Column (2) examines the effect of digital financial inclusion on Internet access, and the digital financial inclusion index is positively significant at the 1 % level. Combined with the theoretical part of the analysis, the increase in Internet access reduces educational inequality (Painter et al., 2022; Agarwal and Canfiel, 2024). Therefore, digital financial inclusion can promote equity in educational opportunities by increasing the Internet penetration rate and thereby. Bootstrap test shows that both direct and indirect effects do not contain 0 at 95 % BC confidence interval, which verifies the significance of the mediation effect. Sobel test ($Z = 3.873$) is also significant, and the mediation effect accounts for 40.3 % of the total. All of these results support the mediating role of Internet penetration in the main effect, and Hypothesis 2 is verified.

4.4. Heterogeneity analysis

Given that there are significant differences in the distribution of economic fundamentals, financial development levels, and educational resources among the central, eastern, and western regions of China, which in turn may affect the mechanism of digital financial inclusion on equity of educational opportunities, this study further divides the provinces into eastern, central, and western regions for heterogeneity analysis. The regression results are shown in Table 4, in the western region, the Findex coefficient is −0.122, which is negatively significant at the 1 % level; while in the eastern region, the Findex coefficient is 0.065, which is positively significant at the 5 % level; and in the central region, the Findex coefficient is 0.033, which fails the significance test.

The above results show that there is significant heterogeneity in the impact of digital financial inclusion on equity of educational opportunities in different regions. This may be because in the western region, where traditional financial services are insufficient, the popularization of digital inclusive finance is more capable of compensating for the shortcomings of traditional financial services, thus playing a more substantial role in narrowing the gap between urban and rural educational resources; whereas in the eastern region, where educational resources and financial services are more adequate, digital finance may be more likely to be prioritized for the use of

Table 4
Heterogeneity analysis results.

	(1)	(2)	(3)
VARIABLES	East	Middle	West
Findex	0.065**	0.033	−0.122***
	(0.032)	(0.035)	(0.036)
Gdp	−0.010*	−0.004	−0.086***
	(0.005)	(0.006)	(0.013)
Reg	0.001**	−0.001	0.007***
	(0.001)	(0.001)	(0.001)
Sav	0.031	0.083***	−0.085**
	(0.029)	(0.027)	(0.034)
Popgrw	0.002***	−0.001	−0.002**
	(0.001)	(0.001)	(0.001)
Inrat	−0.019	−0.005	−0.045**
	(0.014)	(0.017)	(0.020)
Fisc	0.018	0.052**	−0.102**
	(0.015)	(0.023)	(0.042)
Constant	−0.157	0.084	1.118***
	(0.151)	(0.154)	(0.170)
Observations	121	88	132
R-squared	0.546	0.700	0.651
Year FE	YES	YES	YES
ID FE	YES	YES	YES

Note: Standard errors in parentheses.
*, ** and *** indicate significance at the 10 %, 5 % and 1 % levels, respectively.

the higher-income groups, thus exacerbating educational inequality instead.

5. Conclusions

An empirical analysis of panel data from 31 provinces in China from 2013–2023 finds that the development of digital inclusive finance significantly reduces the education Gini coefficient and plays a positive role in promoting equity in education opportunities. Further research finds that digitally inclusive finance realizes the sharing of educational resources by increasing Internet penetration, thus promoting educational equity. In addition, the role of digitally inclusive finance in promoting equity in educational opportunities is also related to regions, and is more significant in the western region, where economic development is relatively lagging and financial services are insufficient.

Based on the above conclusions, this paper has the following suggestions for the development of equity in educational opportunities. First, promote digital urban and rural construction and regional synergistic development. The government should accelerate the construction of informatization, promote the integration of digital technology and industry, narrow the digital divide between urban and rural areas, and create a favorable environment for family investment in education. At the same time, based on the strategy of coordinated economic development, it should increase support for the western region and play a role in driving educational equity. Second, upgrading the level of Internet infrastructure construction in various regions. Provinces have continued to increase investment in Internet infrastructure construction and improve the access rate of Internet education resources. The focus is on promoting network coverage in backward areas and areas with outstanding education equity issues, realizing the full popularization of Internet "physical access", and ensuring that more people can conveniently access high-quality network resources. Third, popularizing financial knowledge and strengthening equity monitoring. Provinces should strengthen the publicity and popularization of digital inclusive financial knowledge and products through online and offline means, to improve the financial literacy of residents. At the same time, they should establish early warning mechanisms for different levels of educational equity and take measures against provinces that experience a rebound in educational equity in the long term and the short term, respectively.

Author statement

The author declares that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

CRediT authorship contribution statement

Meng Zhang: Writing – review & editing, Writing – original draft, Software, Resources, Data curation, Conceptualization.

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Appendix A. Variable definitions

A.1. Explained variable: education Gini coefficient

The Gini coefficient for education (Jini), which quantifies the differences in the distribution of educational opportunities and makes them comparable across regions, has been validated by empirical research, and the data are robust as well as feasible. The value is 0–1, with larger values indicating a higher degree of educational inequality. Specific reference to the method of Luo (2024) to measure the Gini coefficient of education, the formula is as follows:

$$\mathrm{Jini} = \left(\frac{1}{\sigma}\right) \sum_{i=2}^{n} \sum_{j=1}^{i-1} x_i \left(y_i - y_j\right) x_j$$

$$\sigma = \sum_{i=1}^{n} x_i y_i$$

σ represents the average number of years of schooling of the population over 6 years of age in each province;x_i,$y_j$represents the number of years of schooling corresponding to the different levels of education; n represents the number of subgroups at different levels of education, which is set at 5, and divides the population into the following levels according to the level of literacy: illiterate or semi-illiterate (y_1=0), elementary school educated (y_2=6), lower secondary school educated (y_3=9), upper secondary school or middle school educated (y_4=12), and college educated and above (y_5=16).

A.2. Explanatory variables: financial innovation index

Digital Financial Inclusion Index (Findex) adopts the Digital Financial Inclusion Index published by the Digital Finance Research Center of Peking University to measure the level of digital financial development in the region, spanning from 2013 to 2023. It contains three dimensions: the breadth of digital financial coverage, the depth of use, and the degree of digitization of inclusive finance, with 33

specific indicators such as payment, credit, insurance, credit, investment, etc. (Kapoor, 2013). These indicators can reflect the performance of each region in the field of digital financial inclusion more objectively.

A.3. Intermediary variable:Internet Penetration Rate

Internet penetration rate is a core indicator for measuring the level of regional informatization development, reflecting infrastructure coverage and public accessibility. 2013–2016 data are sourced from the National Bureau of Statistics (NBS), and 2017–2022 data are taken from the China Internet Development Report and the 52nd Statistical Report on the Development of the Internet in China, and are measured in terms of the number of broadband access subscribers as a proportion of the province's resident population, To ensure comparability of data across periods. This indicator indirectly drives equity in educational opportunities by influencing the distribution of resources through technological accessibility.

A.4. Control variables.

Gdp stands for gross domestic product, expressed as the logarithm of per capita GDP; Reg stands for the proportion of urban population, calculated as the ratio of the number of urban population to the total population at the end of the year; Sav stands for the savings rate, calculated as the ratio of 1 - per capita consumption expenditure to per capita disposable income; Popgrw stands for the natural population growth rate, equal to the difference between the birth rate and the death rate of the population; Inrat stands for the ratio of urban-rural Income Ratio, measured by the ratio of disposable income per capita for urban residents to disposable income per capita for rural residents; Fics stands for Fiscal Self-sufficiency Ratio, which is equal to the ratio of general public budget revenues to general public budget expenditures.

Data availability

Data will be made available on request.

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